

TASK 3 - SECURE CODING REVIEW

1. SELECT A PROGRAMMING LANGUAGE AND APPLICATION TO AUDIT

I selected Python as the programming language for this secure coding review project.

The application selected is a Simple Login System. The login system checks username and password to allow user access. This application is chosen because login systems commonly contain security vulnerabilities like weak passwords and improper input validation.

2. PERFORM A CODE REVIEW TO IDENTIFY SECURITY

VULNERABILITIES

The code review was performed on the Python Login System to identify security vulnerabilities.

Vulnerability 1 – Hardcoded Password

The password was directly stored inside the code. This is insecure because attackers can easily access the password if they view the source code.

Example:

```
password = "admin123"
```

Vulnerability 2 – Weak Password

Weak passwords are easy to guess or crack. Using simple passwords can lead to unauthorized access.

Vulnerability 3 – No Input Validation

The application does not properly validate user input. Attackers may enter harmful data into the system.

3. USE TOOLS LIKE STATIC ANALYZERS OR MANUAL INSPECTION

METHODS

The secure coding review was conducted using manual inspection methods.

The source code of the Python Login System was carefully reviewed to identify security weaknesses such as hardcoded passwords, weak authentication, and lack of input validation.

Static analysis techniques were also used to examine the code structure and detect possible vulnerabilities without executing the application.

4. PROVIDE RECOMMENTATIONS AND BEST PRACTICES FOR SECURE CODING

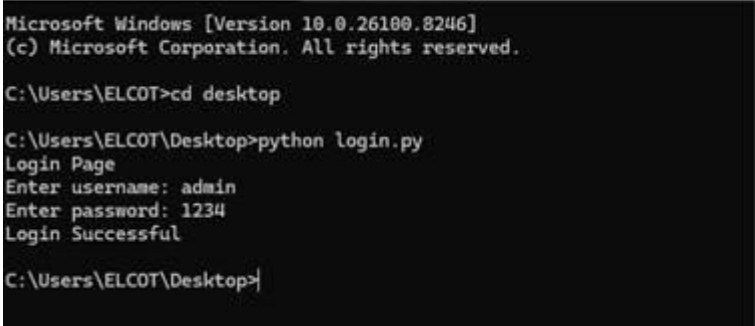
The following recommendations were provided to improve the security of the application:

1. Use strong passwords with letters, numbers, and special characters.
2. Avoid storing passwords directly inside the source code.
3. Validate all user inputs before processing them.
4. Encrypt sensitive information to protect user data.
5. Regularly update software and security patches.
6. Use secure authentication methods to prevent unauthorized access.

These best practices help in developing secure and reliable applications.

EXAMPLE CODE

```
username = input("Enter Username: ")
password = input("Enter Password: ")
if username == "admin" and password == "1234":
    print("Login Successful")
else:
    print("Invalid Login")
```



```
Microsoft Windows [Version 10.0.26100.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ELCOT>cd desktop

C:\Users\ELCOT\Desktop>python login.py
Login Page
Enter username: admin
Enter password: 1234
Login Successful

C:\Users\ELCOT\Desktop>
```